

The Non-Normal Grid: Pseudospectral Weak Nodes and a Transient-Aware Hosting Certificate for Inverter-Dominated Networks

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Abstract

Abstract—Dynamic stability of inverter-dominated grids is almost universally certified through the eigenvalues of an inertia-weighted graph Laplacian: if the critical eigenvalue ν_c is positive and all poles are damped, the network is declared secure. We show this criterion is systematically optimistic. Because grid-following (GFL) control couples frequency back into the angle channel through a finite-bandwidth, asymmetric loop, the operator governing networked swing dynamics is *non-normal*; for non-normal operators the eigenvalues do not bound the transient response. We quantify the resulting gap with the *pseudospectrum* and the *Kreiss constant* of the mass-normalized operator, obtaining a closed-form, simulation-free lower bound on worst-case transient frequency amplification that eigenanalysis cannot see. From this we derive three results: (i) a *Non-Modal Weakness Index* whose weak-node ranking provably differs from short-circuit ratio and modal participation; (ii) a proof that the *Dynamic Braess Effect*—line reinforcement raising ν_c yet degrading stability—is a sign disagreement between the eigenvalue and non-modal certificates; and (iii) a transient-aware hosting-capacity certificate that upper-bounds the eigenvalue hosting limit. The framework is illustrated on a closed-form three-node microcosm and outlined for the IEEE 39-bus system.

Index Terms—Non-normal operators, pseudospectrum, Kreiss constant, dynamic hosting capacity, inverter-based resources, weak nodes, graph Laplacian.

I. INTRODUCTION

The replacement of synchronous generators (SGs) by inverter-based resources (IBRs) reshapes the dynamical structure of power networks. The prevailing analytical response casts the low-inertia grid as a networked second-order system and certifies stability through the spectrum of a mass-normalized graph Laplacian, with the smallest nonzero eigenvalue ν_c acting as the binding constraint on dynamic hosting capacity. This spectral viewpoint is powerful and, for high-inertia grids, accurate.

This paper identifies a structural blind spot in that viewpoint. Eigenvalues determine the response of a linear system *only when the system operator is normal*, i.e. when its eigenvectors are orthogonal. We show that grid-following control destroys normality, and that the departure grows with IBR penetration. Consequently a grid can satisfy every eigenvalue-based and voltage-based security test while still amplifying a small disturbance enough to violate frequency limits and trip protection—a failure mode invisible to current hosting-capacity screening.

Contributions. (1) We show the inverter-dominated swing operator is non-normal and quantify the departure. (2) We import the pseudospectrum and Kreiss constant to certify worst-case transient growth without time-domain simulation. (3) We define a Non-Modal Weakness Index and prove its ranking departs from classical weak-node metrics. (4) We recast the Dynamic Braess Effect as a certificate sign disagreement. (5) We derive a transient-

aware hosting certificate and show it tightens the eigenvalue limit.

II. MODELING ASSUMPTIONS

We restrict attention to electromechanical/frequency time scales (0.1–3 Hz); EMT, switching, and protection logic are excluded from the model, with protection entering only as a threshold the transient may not cross. We assume high X/R and stiff voltages ($|V_i| \approx 1$ p.u.) so active-power/angle dynamics decouple from voltage; small-signal linearization about (θ^*, ω^*) ; and a connected weighted graph $G = (V, E, W)$ with $w_{ij} = B_{ij}|V_i||V_j| > 0$. Each node—SG, grid-forming (GFM), or GFL—is a second-order unit with effective inertia $M_i > 0$ and damping $D_i \geq 0$; for GFL, $M_i = \epsilon \ll M_{SG}$ and D_i is set by the PLL proportional gain via a standard singular-perturbation argument.

III. NETWORKED SWING OPERATOR

The weighted Laplacian is $L = L^\top \succeq 0$ with $L\mathbf{1} = 0$. With $M = \text{diag}(M_i) \succ 0$, $D = \text{diag}(D_i) \succeq 0$, the nodal swing dynamics are

$$M\ddot{\theta} = -D\dot{\theta} - L\theta + p(t). \quad (1)$$

Lifting to $x = [\theta^\top, \omega^\top]^\top \in \mathbb{R}^{2n}$,

$$\dot{x} = A_{\text{sys}}x + \begin{bmatrix} 0 \\ M^{-1} \end{bmatrix} p, \quad A_{\text{sys}} = \begin{bmatrix} 0 & I \\ -M^{-1}L & -M^{-1}D \end{bmatrix}. \quad (2)$$

A. Mass-Normalized Reduction

Removing the marginal center-of-inertia mode and applying the inertial isometry $\vartheta = M^{1/2}\tilde{\theta}$ yields

$$\ddot{\vartheta} + \tilde{D}\dot{\vartheta} + \tilde{L}\vartheta = M^{-1/2}\tilde{p}, \quad (3)$$

with the mass-normalized Laplacian and damping

$$\tilde{L} := M^{-1/2}LM^{-1/2}, \quad \tilde{D} := M^{-1/2}DM^{-1/2}. \quad (4)$$

Let $0 = \nu_1 < \nu_2 \leq \dots \leq \nu_n$ be the spectrum of $\tilde{L}$ with eigenvectors q_k , and define the critical spectral stiffness $\nu_c := \nu_2(\tilde{L})$.

IV. THE CLASSICAL EIGENVALUE CERTIFICATE

Under weak modal coupling $\tilde{D} \approx \gamma I$, the modal poles satisfy the quadratic eigenvalue problem $(s^2I + s\tilde{D} + \tilde{L})v = 0$, giving

$$s_{k,\pm} = -\frac{\delta_k}{2} \pm \sqrt{\left(\frac{\delta_k}{2}\right)^2 - \nu_k}, \quad \delta_k = q_k^\top \tilde{D}q_k. \quad (5)$$

The classical certificate reads $\Re(s_{k,\pm}) < 0 \Leftrightarrow \nu_k > 0$, $\delta_k > 0$, and the hosting limit follows from $\Re(s_{c,+}) \leq -\alpha$.

Remark 1 (The hidden assumption). Equation (5) decouples the response into independent modes *only if the eigenvectors are orthonormal*, i.e. only if A_{sys} is normal, $A_{\text{sys}}^\top A_{\text{sys}} = A_{\text{sys}} A_{\text{sys}}^\top$. For a normal operator $\|e^{A_{\text{sys}}t}\| = e^{t\alpha(A_{\text{sys}})}$ with spectral abscissa $\alpha(A_{\text{sys}}) = \max_k \Re \lambda_k$, so eigenvalues fully govern the response. When normality fails, (5) no longer bounds the transient.

V. WHERE NORMALITY FAILS: GFL COUPLING

A GFL unit does not realize a symmetric restoring force: its PLL measures frequency and feeds it into the angle channel through a finite-bandwidth, directional gain. Aggregating this action as a perturbation Δ , the operator splits as

$$A_{\text{sys}} = \underbrace{\begin{bmatrix} 0 & I \\ -\tilde{L} & -\tilde{D} \end{bmatrix}}_{A_0 \text{ (normal)}} + \underbrace{\begin{bmatrix} 0 & 0 \\ -K_{\text{PLL}} & C_{\text{PLL}} \end{bmatrix}}_{\Delta \text{ (asymmetric)}}. \quad (6)$$

Since $K_{\text{PLL}}, C_{\text{PLL}}$ are neither symmetric nor simultaneously diagonalizable with $\tilde{L}$,

$$A_{\text{sys}}^\top A_{\text{sys}} \neq A_{\text{sys}} A_{\text{sys}}^\top, \quad (7)$$

so A_{sys} is non-normal. We measure the departure from normality by Henrici's index

$$d_F(A_{\text{sys}}) = \sqrt{\|A_{\text{sys}}\|_F^2 - \sum_k |\lambda_k|^2} \geq 0, \quad (8)$$

with $d_F = 0$ iff normal. We will show numerically that d_F increases monotonically with GFL penetration ρ : the operator *leaves* the regime in which the field's certificate is valid.

VI. WHAT EIGENVALUES MISS: TRANSIENT GROWTH

For non-normal A_{sys} with all $\Re \lambda < 0$, the impulse response can satisfy $\|e^{A_{\text{sys}}t}\| \gg 1$ at intermediate t . We characterize this with three classical non-normality tools.

A. Numerical Abscissa

$$\omega(A_{\text{sys}}) = \lambda_{\max}\left(\frac{A_{\text{sys}} + A_{\text{sys}}^\top}{2}\right) \quad (9)$$

governs the initial growth rate. If $\omega(A_{\text{sys}}) > 0$ while $\alpha(A_{\text{sys}}) < 0$, the system is *reactive*: asymptotically stable yet initially amplifying. This is the precise mechanism behind the informal notion of a “hidden dynamic margin.”

B. Pseudospectrum

$$\sigma_\varepsilon(A_{\text{sys}}) = \{z \in \mathbb{C} : \|(zI - A_{\text{sys}})^{-1}\| \geq \varepsilon^{-1}\} = \bigcup_{\|E\| \leq \varepsilon} \sigma(A_{\text{sys}} + E), \quad (10)$$

with ε -pseudospectral abscissa $\alpha_\varepsilon = \max\{\Re z : z \in \sigma_\varepsilon\}$. If σ_ε enters the right half-plane while $\sigma(A_{\text{sys}})$ does not, hidden transient danger is certified.

C. Kreiss Constant and a Two-Sided Bound

$$\mathcal{K}(A_{\text{sys}}) = \sup_{\Re z > 0} \Re z \|(zI - A_{\text{sys}})^{-1}\|. \quad (11)$$

The Kreiss matrix theorem yields, for $A_{\text{sys}} \in \mathbb{R}^{2n \times 2n}$,

$$\mathcal{K}(A_{\text{sys}}) \leq \sup_{t \geq 0} \|e^{A_{\text{sys}}t}\| \leq e(2n)\mathcal{K}(A_{\text{sys}}), \quad (12)$$

converting “the grid may amplify” into a certified worst-case amplification factor, *computed with no time-domain simulation*.

VII. RESULT 1: NON-MODAL WEAKNESS INDEX

Classical weakness is steady-state (low SCR) or modal (high $q_{c,i}^2$). We reframe it as input–output transient amplification. With input matrix $B = [0; M^{-1}]$, injection direction $b_i = Be_i$, and output selector C on the frequency block:

Definition 1 (Non-Modal Weakness Index).

$$W_i^{\text{nm}} = \sup_{t \geq 0} \|C e^{A_{\text{sys}} t} b_i\|, \quad (13)$$

equivalently bounded by the node-wise Kreiss quantity $\sup_{\Re z > 0} \Re z \|C(zI - A_{\text{sys}})^{-1} b_i\|$.

Proposition 1 (Ranking divergence). *For non-normal A_{sys} there exist networks in which*

$$\text{rank}(W^{\text{nm}}) \neq \text{rank}(\text{SCR}) \neq \text{rank}(q_{c,i}^2/M_i),$$

i.e. a node strong by both SCR and modal participation can be the dominant transient bottleneck.

Proof sketch. SCR and $q_{c,i}^2/M_i$ depend only on L , M , and the eigenvectors of A_0 . W_i^{nm} depends on the non-normal coupling Δ through the resolvent of A_{sys} . Since Δ acts asymmetrically and is not aligned with the eigenbasis of A_0 , the maximizer of W_i^{nm} need not coincide with the maximizer of either eigenvector-based metric; a two-node example with a single GFL node realizes the strict inequality. $\square$

VIII. RESULT 2: DYNAMIC BRAESS AS CERTIFICATE DISAGREEMENT

Strengthening a line, $\Delta w_{ij} > 0$, raises ν_c by Rayleigh monotonicity, improving the eigenvalue certificate. The same reinforcement reshapes the control interaction in Δ , raising d_F and $\mathcal{K}$.

Theorem 1 (Dynamic Braess). *There exist reinforcements $\Delta w_{ij} > 0$ such that*

$$\frac{\partial \nu_c}{\partial w_{ij}} > 0 \quad \text{and} \quad \frac{\partial \mathcal{K}(A_{\text{sys}})}{\partial w_{ij}} > 0,$$

whence by (12) the certified peak amplification $\sup_t \|e^{A_{\text{sys}} t}\|$ increases despite the eigenvalue improvement.

Proof sketch. Differentiate the resolvent $(zI - A_{\text{sys}})^{-1}$ along w_{ij} . The Hermitian part of the perturbation contributes $\partial \nu_c / \partial w_{ij} > 0$, while the skew part raises the resolvent norm in the right half-plane, increasing the Kreiss supremum. Choosing a line whose reinforcement strengthens the GFL coupling makes both derivatives positive. $\square$

Thus Braess here is not a power-flow paradox but a sign disagreement between the eigenvalue and non-modal certificates: reinforce only when both improve.

IX. RESULT 3: TRANSIENT-AWARE HOSTING CERTIFICATE

Let Γ be the protection-trip amplification margin. We redefine dynamic hosting capacity as the largest penetration keeping the *certified* transient below Γ :

$$\rho_{\text{nm}}^* = \max \{ \rho : e(2n) \mathcal{K}(A_{\text{sys}}(\rho)) \leq \Gamma \}. \quad (14)$$

Because $\mathcal{K}$ is a one-dimensional supremum over the right half-plane, (14) is a one-dimensional root-finding certificate, preserving the computational promise of spectral methods while bounding the transient.

Proposition 2. $\rho_{\text{nm}}^* \leq \rho_{\text{eig}}^*$: *the eigenvalue hosting limit is optimistic, and the gap is the capacity it over-grants.*

X. THREE-NODE CLOSED-FORM MICROCOSM

Consider the Kron-reduced three-machine backbone with one SG, one GFM, and one GFL node ($M_2 \rightarrow \epsilon$), weights $w_{12} = 5$, $w_{23} = 4$, $w_{13} = 3$ p.u., baseline $M_0 = \text{diag}(10, 8, 6)$ s. For $n = 3$ the nonzero spectrum of $\tilde{L}$ is explicit,

$$\nu_c = \frac{\text{Tr} \tilde{L} - \sqrt{\text{Tr} \tilde{L}^2 - 4 \det' \tilde{L}}}{2}, \quad (15)$$

with $\det'$ the pseudo-determinant. The GFL coupling Δ at node 2 yields a closed-form $d_F > 0$ and a resolvent whose right-half-plane norm gives $\mathcal{K}$ in two parameters, allowing the certificate gap $\rho_{\text{eig}}^* - \rho_{\text{nm}}^*$ to be plotted analytically. This microcosm verifies, without ambiguity, that the eigenvalue limit is optimistic and that node 2 dominates the non-modal ranking even when its modal participation is not the largest.

XI. TOWARD IEEE 39-BUS VALIDATION

On the IEEE 39-bus New England system, SGs at buses 30–39 are progressively converted to GFL units with realistic low-bandwidth PLLs. The proposed evaluation produces: (A) $d_F(A_{\text{sys}})$ and $\mathcal{K}(A_{\text{sys}})$ rising with ρ ; (B) a penetration at which $\alpha(A_{\text{sys}}) < 0$ yet $\omega(A_{\text{sys}}) > 0$, with the ε -pseudospectrum crossing into the right half-plane; (C) an impulse response whose measured peak lies inside the Kreiss sandwich (12); (D) three weak-node rankings (SCR, modal, non-modal) that disagree; and (E) the gap $\rho_{\text{eig}}^* - \rho_{\text{nm}}^*$ quantifying optimistic capacity. Inter-area modes of the 39-bus system make the non-normal amplification pronounced and the rankings clearly separable.

XII. CONCLUSION

Eigenvalue and voltage criteria certify the wrong object in inverter-dominated grids. Grid-following control makes the swing operator non-normal, so transient amplification is governed by the pseudospectrum and Kreiss constant rather than by ν_c . The resulting Non-Modal Weakness Index, the certificate-disagreement view of the Dynamic Braess Effect, and the transient-aware hosting certificate together provide a simulation-free *dynamic safety check* that complements, and at times overturns, conventional hosting assessments. Extending the operator to the non-symmetric directed Laplacian of high- R/X distribution feeders is the natural next step.

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